

Propositional Calculus

Definition: a proposition is a statement that is either true or false

e.g., $1 \neq 2$, all cats are black

Definition: a predicate is a statement involving one or more free variables, which become a proposition when we specify values for the free variables.

Quantifiers

$\forall$ = "for all"

$\exists$ = "there exists"

Operations on propositions

$\vee$ = "or" $\Rightarrow$ = "implies"

$\wedge$ = "and" $\Leftarrow$ = "implied by"

$\neg$ = "not" $\Leftrightarrow$ = "if and only if" / "iff"